Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **BreederNear AI**
b. Team leader name: **Sudarsan N**
c. Problem Statement: **Retail & Commerce: trusted, legal breeder-to-buyer pet commerce in India (BreederNear AI)**

Brief about the idea

BreederNear AI: find the trusted breeder next door.

The problem

- India has about 39 million pets[1], but buying one is a gamble. OLX stopped live-animal listings[2], so the trade moved to WhatsApp and Facebook groups with no checks at all.
- Buyers overpay in shops or lose money to “full advance, courier only” posts with stolen photos.
- Protected native birds are still sold: 300+ of India’s ~1,200 bird species are found in illegal trade[3], and munias are dyed to disguise them[4].
- Real local breeders, often a street away, are invisible and lose hours to manual selling.

Our solution

- A breeder-to-buyer pet marketplace where every listing is screened by Gemini and rules in code before any buyer sees it, and breeders can earn a Verified badge.
- Sellers list from photos and a casual Tamil or English message in under a minute.
- Buyers see a match % for their home, book a visit, pay in person, then get a care companion for the pet at home.
- “Is this post safe?”: buyers can check any post they saw in a WhatsApp or Instagram group.

Built from our founder’s five years as a bird breeder and broker in Coimbatore.

[1] Euromonitor, Pet Care in India (2025) · [2] OLX India: items not allowed · [3] WWF-India / TRAFFIC · [4] Wildlife SOS, Delhi seizure report

Opportunities

How is it different?

- Classifieds such as OLX stopped live-animal listings; big pet platforms sell food and accessories.
- A few kennel websites sell dogs, checking only themselves.

No platform we found checks every listing for wildlife law, registration, scam prices and reused photos before it goes live, or checks posts from outside groups.

How does it solve the problem?

- Gemini turns photos and a messy message into a structured, fairly priced listing.
- Trust is decided by transparent code: protected species blocked, dog registration checked, price and scam-language warnings, reused-photo detection.
- Buyers see why: every badge lists its reasons and the questions to ask.
- In-person handover; AI-drafted replies; starter kit and care plan.

USP

Trust built in, not bolted on.

- Protected species blocked from the photo alone: 8/8 in testing, 0/37 false alarms.
- The AI can't override the rules: prompt injection still ends in CAUTION (eval E12).
- Works where the trade happens: Tamil/English messages and outside posts.
- One tool layer for tap screens and the AI chat.

List of features offered by the solution

For buyers (customer app)

- Pets near me with a match % for your home, trusted first, fair-price bar
- Listing page: trust checks, health notes, questions to ask
- AI-drafted first message; scam warnings on replies
- Book a visit; “I bought this pet” after it (paid in person)
- My pets: daily log with AI feedback, growth, health calendar
- Symptom checker: how urgently to see a vet (never medicines)
- “Is this post safe?”, starter kit, Local Breeders directory

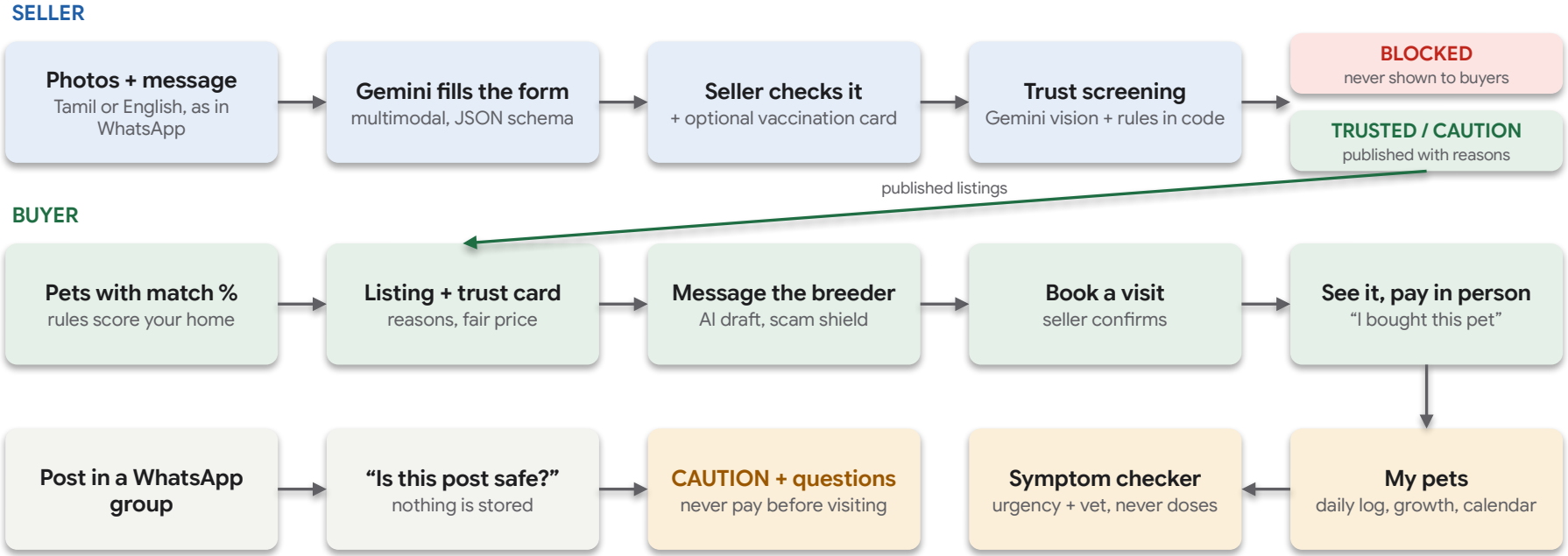
For sellers (seller app)

- “Fill with AI”: photos + a Tamil/English message become a listing
- Publish with trust check: TRUSTED, CAUTION or BLOCKED, with reasons
- Dashboard charts: views, enquiries, visits, sales, funnel
- Verified breeder badge: phone code, facility photos read by Gemini, dog-breeder certificate (no ID documents)
- Visit requests to confirm; AI-drafted replies to enquiries
- Vaccination card read into the trust check

Everywhere

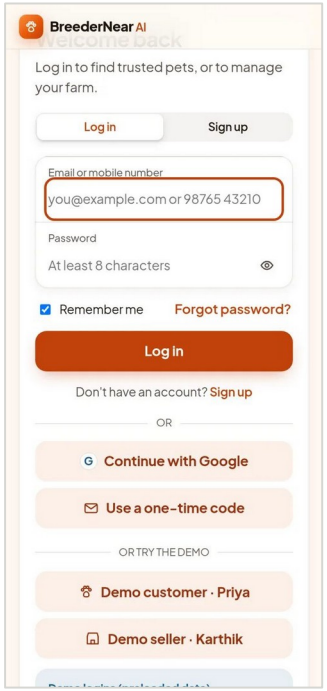
Seven Gemini agents in English or Tamil use the same tools as the screens. **Sign-in:** password, one-time code or Google (Firebase Authentication). **Safety:** roles checked in the API and the AI tools, rate limits, every AI reply checked for medicine doses. **Accessible:** WCAG 2.2 AA, phone to desktop.

Process flow diagram or Use-case diagram

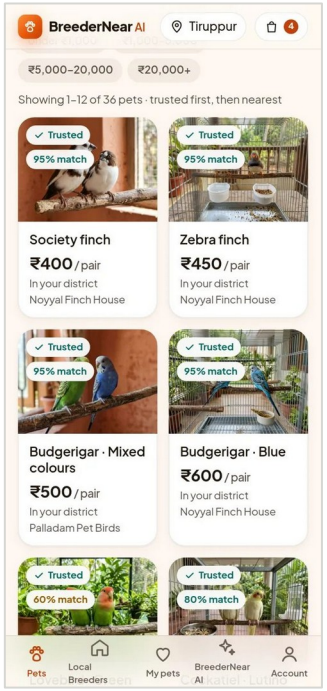


Every step works by tapping, or by asking BreederNear AI in chat: the same tools, the same checks. Sellers also earn a Verified breeder badge.

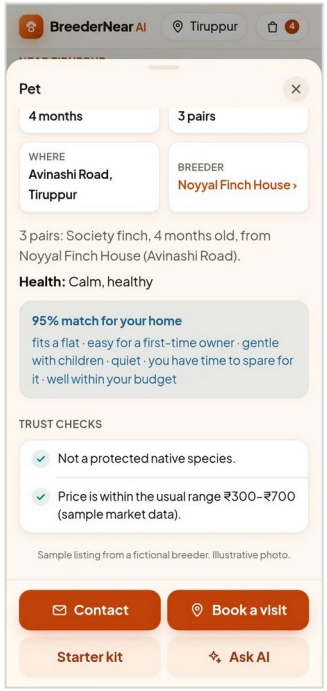
Wireframes/Mock diagrams of the proposed solution (optional)



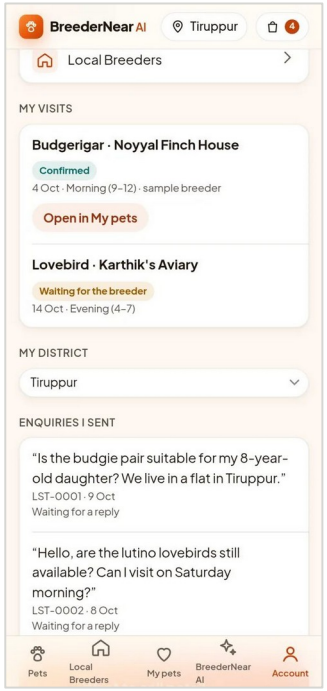
Sign in or one-tap demo



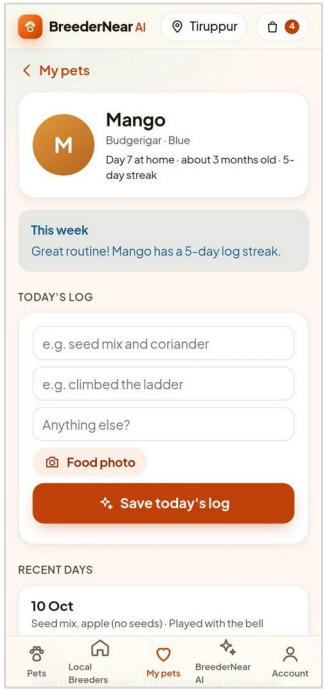
Pets with match %



Listing + trust checks

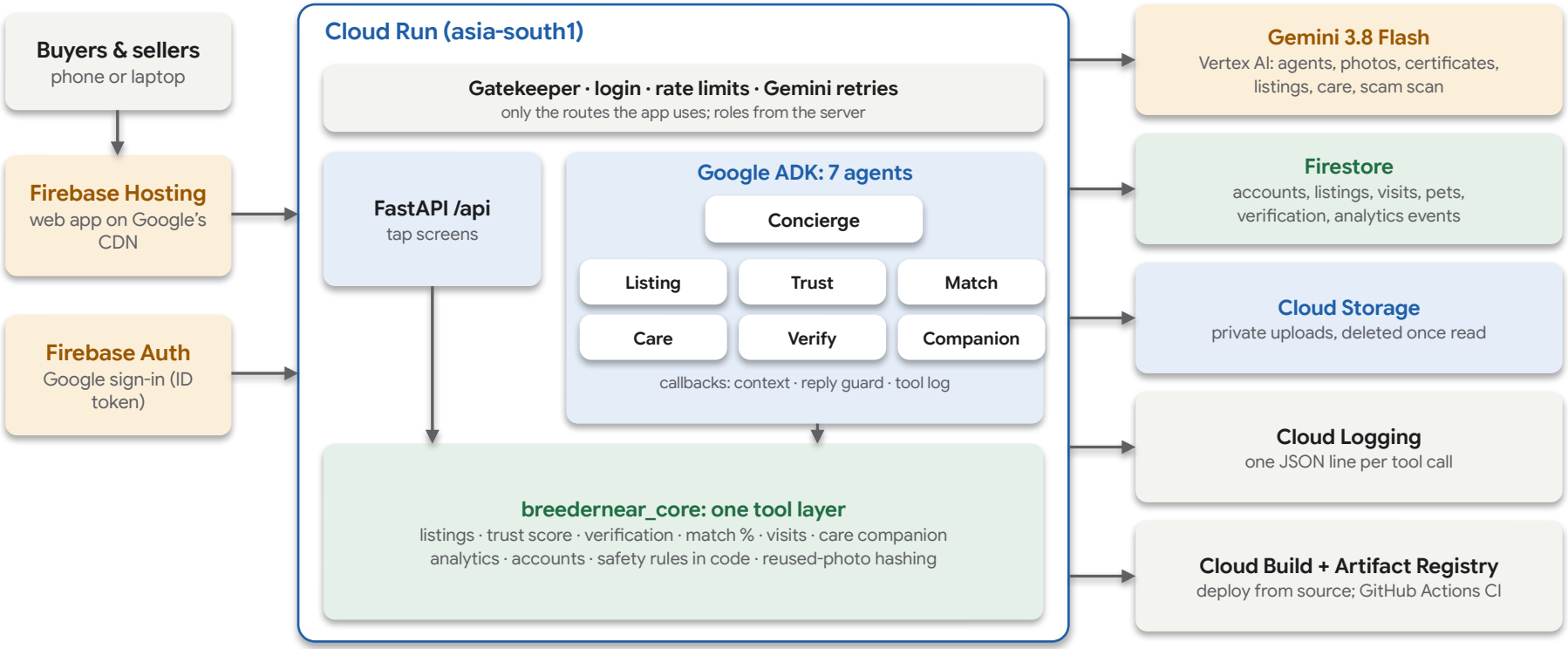


Visits, then "I bought it"



My pets companion

Architecture diagram of the proposed solution



Technologies to be used in the solution

Google technology	How BreederNear AI uses it
Gemini 3.8 Flash (Vertex AI)	Seven ADK agents; listing extraction from photos + Tamil/English; photo, facility and certificate reading; care plans, daily-log feedback, symptom urgency; message drafts and scam scan. JSON schemas.
Google ADK (Python)	Concierge + listing, trust, match, care, verification and companion agents; tools; callbacks on every agent; streamed replies
Cloud Run	API and agents; login gatekeeper; rate limits; Gemini retries with backoff
Firebase Hosting	The web app on Google's CDN; calls Cloud Run directly (CORS) so AI replies stream
Firebase Authentication	Continue with Google; the server verifies the Firebase ID token
Firestore	Accounts, listings, enquiries, visits, sales, pets and logs, verification, analytics
Cloud Storage, Cloud Build, Artifact Registry, Cloud Logging	Private uploads; deploy from source; tool-call logs
gemini-3.1-flash-image	Illustrative photos for the fictional sample listings (labelled in the app)

Also used

- Python 3.12, FastAPI, Pydantic
- imagehash: reused-photo detection
- Plain HTML, CSS, JavaScript and SVG charts: no build step
- pytest, Playwright + axe-core, GitHub Actions
- Firebase CLI (deploy only)

Estimated implementation cost (optional)

AI action (measured, real calls)	Gemini tokens (in · out incl. thinking)	Cost per action*
List a pet: extract listing + screen photo	≈ 4,560 · ≤ 1,075	≈ ₹1.3
Check an outside post (text only; a screenshot adds ≈ ₹0.5)	≈ 1,620 · ≤ 445	≈ ₹0.5
Verification: read a certificate · review 2 facility photos	≈ 1,640 · 400 · ≈ 2,630 · 645	≈ ₹0.5 · ≈ ₹0.8
Buyer profile from their words · scan a message	≈ 700 · 550 · ≈ 300 · 120	≈ ₹0.5 · ≈ ₹0.1
14-day care plan (cached per species and age) · reply draft	≈ 560 · ≤ 1,680 · ≈ 500 · ≤ 790	≈ ₹1.2 · ≈ ₹0.6

Running cost	Prototype through judging	Example: 10,000 listings a month
Cloud Run (1 always-on instance)	≈ ₹600–1,000 / month (estimate)	scales per request
Firestore, Cloud Storage, Firebase Hosting + Auth	within free usage at prototype scale	pay per use (not measured)
Gemini	a few hundred ₹ (tests + judges)	≈ ₹13,000 / month for screening

Screening a listing costs about ₹1.3 against animals priced ₹400–₹25,000 in our sample listings, so every listing can be checked.

*Measured tokens (10–11 Oct 2026); thinking tokens counted as output · Gemini 3.8 Flash at $1.50 in / $7.50 out per 1M tokens (list price from 1 Jan 2027 as reported by price trackers; introductory price until then is half) · ₹88 = $1 (assumption) · Estimates are labelled

Business case and go-to-market

Revenue (planned)

- Verified-breeder subscription: badge, dashboard insights, more listings
- Featured listings in a district
- Commission on starter kits and accessories from partner brands
- B2B: trust-screening API for classifieds and pet shops

Why it works

- Screening costs about ₹1.3 per listing (measured tokens)
- Serverless: Cloud Run and Firestore cost little when idle and scale per district
- Trust is the product: OLX left this market because it couldn't police sellers

Go-to-market

- Start where the founder trades: Coimbatore and Tiruppur breeder groups
- Breeders onboard with one message and their phone photos
- Buyers arrive via "Is this post safe?" links shared in the same groups
- Then district by district across Tamil Nadu

Who benefits

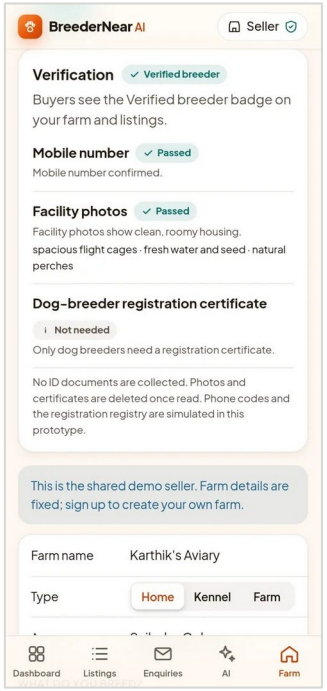
Buyers: fair prices and healthier, legal pets. Breeders: hours saved and buyers beyond their own contacts. Wildlife: fewer protected birds traded, because illegal listings never go live.

Revenue lines and the rollout are plans and assumptions, not results.

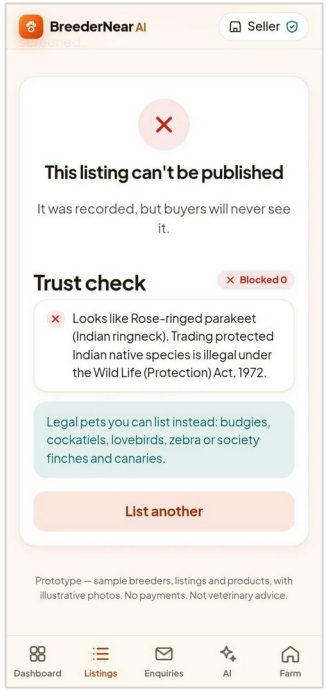
Snapshots of the prototype



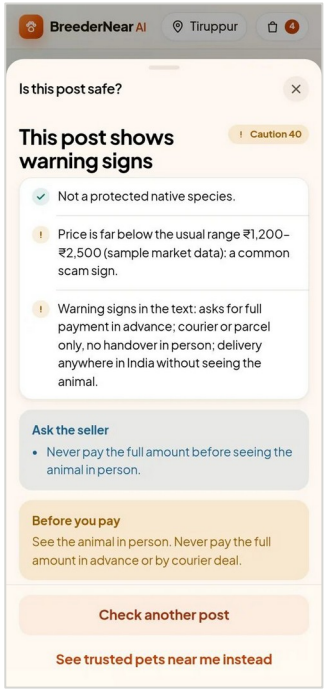
Seller analytics



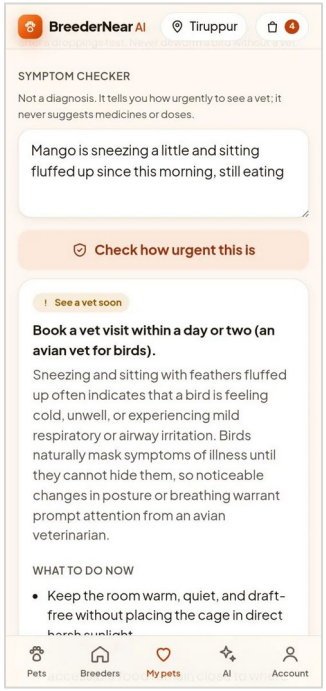
Verified breeder



Protected species: blocked



"Is this post safe?"



Symptom urgency, no doses

Prototype Performance report/Benchmarking

Quality (real Gemini, 10–11 Oct 2026)	Result
Agent evals: 22 cases × 3 runs (listing, Tamil, blocking, scams, reused photos, prompt injection, no medicine advice, roles, verification, visits, care companion)	**66/66 passed**
Protected species blocked from the photo alone	**8/8**
Ordinary pets wrongly flagged as protected	**0/37**
All photo checks: species, dye, blur, no animal, watermark, screenshot	**202/202***
Accessibility: axe-core, WCAG 2.2 AA, light + dark, 320–1280 px, all screens	**0 problems**
Unit tests in CI · live smoke test	**317 · 17/17**

Speed (live)	Time
List a pet: upload, AI fill, publish with trust check	17–27 s
Check a pasted post	5–7 s
Web app load (Firebase CDN)	≈ 0.7 s
Gemini: read a certificate · scan a message	≈ 4.5 s · ≈ 2 s
Gemini: symptom urgency · daily-log feedback	≈ 8 s

How we test

Gemini isn't deterministic, so each eval runs three times and we report the count. Decisions (trust, verification, match %, urgency floor) are made by tested code.

*Test photos are AI-generated and the screening prompt was tuned on them (first run 196/202), so treat this as an upper bound · Reproduce: scripts/run_evals.sh 3 · scripts/vision_accuracy.py · pytest -m ui tests/ui · evals/RESULTS.md, evals/VISION_RESULTS.md

Additional Details/Future Development (if any)

Roadmap (not in the prototype)

- Pilot with real breeders in Coimbatore and Tiruppur, from the founder's network
- Real SMS and email delivery for one-time codes
- List by sending a WhatsApp message (WhatsApp Business Platform)
- Live State Animal Welfare Board and PARIVESH registry checks; the platform registrations India's pet-trade rules require
- Chat history in Agent Engine sessions, so Cloud Run can scale past one instance
- Breeder ratings from buyers' care-companion outcomes; vet and pet-pharmacy partners; more languages

Responsible by design (already built)

- Protected species blocked in code, whatever the model says
- No medicine names or doses: checked on every AI reply, even while streaming
- Verification without ID documents or selfies; photos deleted once read
- Codes shown on screen only open new or demo accounts, never someone else's
- Symptom checker: red flags are always an emergency; never a diagnosis
- Synthetic sample data, clearly labelled; simulated registry numbers start with SIM-

Provide links to your:

- **GitHub Public Repository**: https://github.com/sudarsan-22/breedernear-ai
- **Demo Video Link (3 Minutes)**: [add the YouTube link on 17 Oct]
- **Final Product Link**: https://breedernear-ai-2026.web.app (API: https://breedernear-655711985039.asia-south1.run.app)

Try it in 2 minutes: demo accounts with preloaded data

Customer: priya.customer@example.com **Seller:** karthik.seller@example.com **Password:** demo12345

Or tap “Demo customer · Priya” / “Demo seller · Karthik” on the login screen. The README has a step-by-step walkthrough.

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Thank you

